Introduction To Algorithms

Week 5 – **Recursion, Iteration, and Strategy**

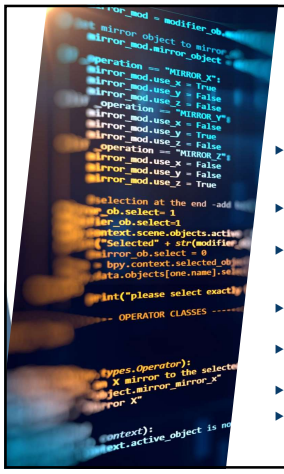
"Same Problem, Different Strategy"

1

Terms to Remember

Term	Brief Meaning
Strategy	A general approach for solving a problem
Correctness	Whether the algorithm produces the expected result
Performance	How efficiently the algorithm uses work or resources
Scalability	How the approach behaves as size or demand grows
Elasticity	Ability to handle changing demand or scale when resources change
Polynomial	A growth pattern that can be expressed with powers such as n, n2, or n3
NP-Hard	A class of problems that are at least as hard as the hardest problems in NP
NP-Complete	Problems that are both in NP and as hard as any problem in NP
Divide and conquer	Split, solve smaller pieces, then combine
Dynamic programming	Store and reuse results from overlapping subproblems
Greedy strategy	Make the best-looking choice at each step
Recursion	A function solves a smaller version of the same problem
Base case	The condition where recursion stops

2



What We Will Use Today

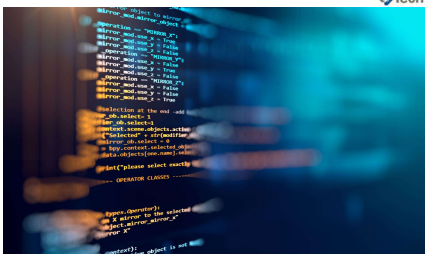
- Correctness, performance, and scalability as strategy questions
- Divide and conquer as a split-and-combine pattern
- Dynamic programming as recognition of saved repeated work
- Greedy as local choice with possible risk
- Iteration and recursion as two strategy forms
- Comparison table evidence
- Strategy recommendation with conditions

3

What We Will "Touch On" and Revisit Later

- Graphs and traversal strategy
- Recommendation and ranking systems
- Larger-scale algorithms
- Explainability and limits of algorithmic solutions
- Final-assessment explanation and justification

4



Review Coding Lab

5

Review: What Lab 04 Taught Us

Last week, we learned that:

Search strategy depended on assumption

Binary search was powerful only when:

- The data was sorted
- The comparison rule matched the order
- The trace decisions were valid

6

Today's Question

How can two correct solutions still be different in quality?

7

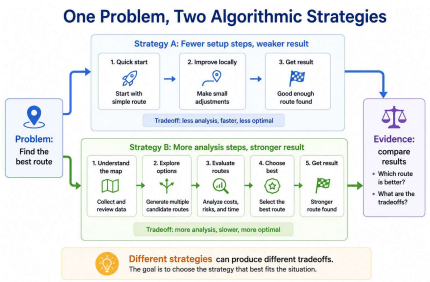
Same Problem, Different Strategy

Many problems can be solved in more than one way.

Strategies may differ in:

- how they break down the problem
- how much work they repeat
- how easy they are to explain
- how well they fit the data shape

8



Strategy Comparison

9

What Counts as Success Today

- ***Describe*** two strategies
- ***Test*** both strategies
- ***Compare*** correctness and readability
- ***Discuss*** likely growth behavior
- ***Recommend*** one strategy with conditions

10

Textbook Review

Designing Algorithms

The reading focuses on designing algorithms.

Design means choosing a strategy that fits:

- The required result
- The problem constraints
- The data size
- The cost of the approach
- The explanation needed

11

Textbook Review

Functional Requirements

Functional requirements ask:

- What must the algorithm produce?
- What input does it need?
- What output is expected?
- What counts as a correct result?

12

Textbook Review

Non-Functional Requirements

Non-functional requirements ask:

- How fast should it be?
- How readable should it be?
- How much memory can it use?
- How well should it scale?
- How explainable does it need to be?

13

Textbook Review

Three Design Concerns

When designing an algorithm, ask:

1. Is it correct?
2. Is the performance reasonable?
3. Can it scale when the data or demand grows?

14

Three Algorithm Design Concerns

1. Correctness	2. Performance	3. Scalability
Does it produce the expected result?	How much work does it take?	What happens as the data grows?

Three Design Concerns

15

Textbook Review

Theory Vocabulary

The reading includes skim-level vocabulary:

- polynomial
- NP-Hard
- NP-Complete

For this course:

- Recognize the terms
- Understand why some problems are harder
- Do ***not*** try to master formal proof

16

Textbook Review

Strategy Families

The reading introduces strategy families:

- divide and conquer
- dynamic programming
- greedy strategy

It also presents larger examples:

- PageRank
- linear programming

17

Design Concerns

18

Correctness

Correctness asks:

- Did the algorithm produce the expected result?
- Did it handle normal cases?
- Did it handle edge cases?
- Can we show evidence?

19

Performance

- How much work does the strategy do?
- Does it repeat unnecessary work?
- How does the work grow?
- Is this approach reasonable for the problem size?

20

Scalability

- What happens with larger data?
- What happens with more users?
- What happens with frequent updates?
- Can the approach adapt when demand changes?

21

Strategy Comparison Frame

Criterion	Strategy A	Strategy B
Correctness		
Readability		
Growth		
Fit to data		

22

Better Means Better For This Context

Avoid

- "This strategy is better."

Use

- "This strategy is better for this problem because..."
- - "This strategy may not be better when..."

23

Evidence For Strategy Choice

Evidence may include:

- test cases
- edge cases
- trace tables
- decision trees
- timing results
- comparison tables
- limitation notes

24

Theory Orientation

25

Polynomial, In Plain Language

Polynomial growth uses powers of input size.

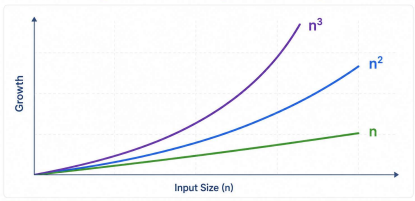
Examples:

- n
- n2
- n3

For now, connect it to growth vocabulary, not formal proof.

26

Recognize the shape, do not panic.



Polynomial usually means growth like n, n^2, or n^3.

Polynomial Recognition

27

NP-Hard And NP-Complete: *Recognition Only*

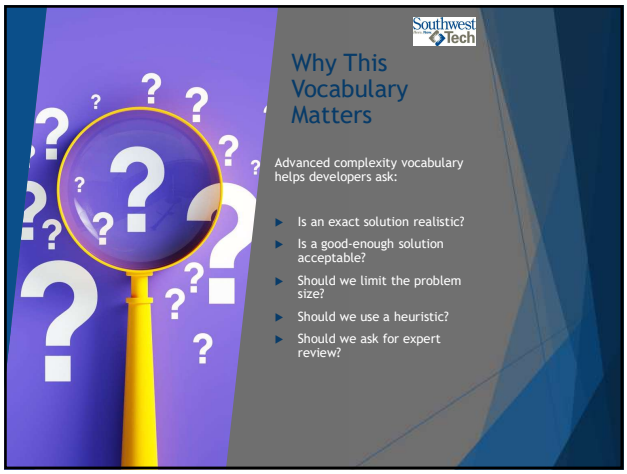
For this course:

- NP-Hard means very hard problem family
- NP-Complete means a special class of hard decision problems
- exact definitions require more theory

Main takeaway:

- *Some problems become hard very quickly!*

28



Why This Vocabulary Matters

Advanced complexity vocabulary helps developers ask:

- Is an exact solution realistic?
- Is a good-enough solution acceptable?
- Should we limit the problem size?
- Should we use a heuristic?
- Should we ask for expert review?

29

You are not expected to master formal complexity theory today.

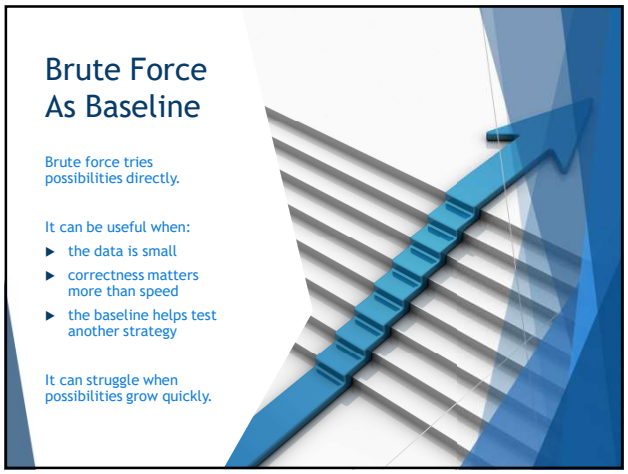
You are expected to:

- recognize that problem difficulty varies
- connect strategy choice to constraints
- explain tradeoffs honestly
- use AI and references carefully

Don't Panic Over Theory

30

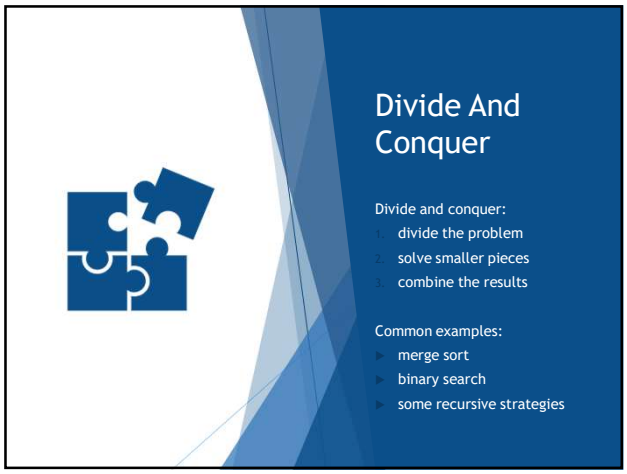
31



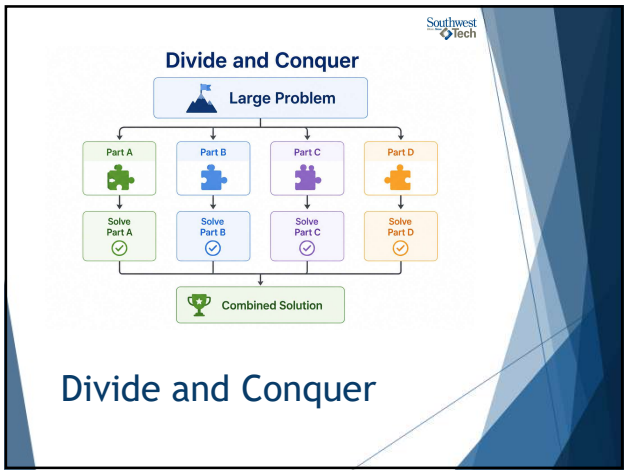
32

33

34



35



36

Divide And Conquer Fit

Divide and conquer fits when:

- the problem can be split
- smaller pieces are similar to the whole
- results can be combined
- splitting reduces the work or complexity

37

Dynamic Programming

Dynamic programming is useful when:

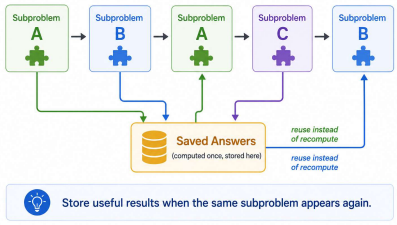
- the problem has repeated subproblems
- earlier results can be saved
- saved results prevent repeated work

For this course: recognize the pattern.

38

Dynamic Programming

Store and reuse repeated subproblem answers.



Dynamic Programming

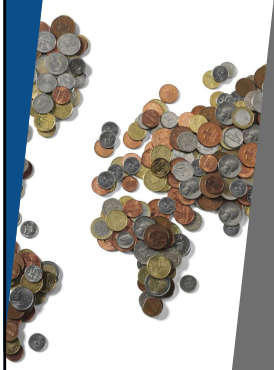
39

Greedy Strategy

A greedy strategy chooses what looks best right now.

Examples:

- choose the cheapest item first
- choose the shortest task first
- choose the largest coin first
- choose the highest immediate score



40

Greedy Risk

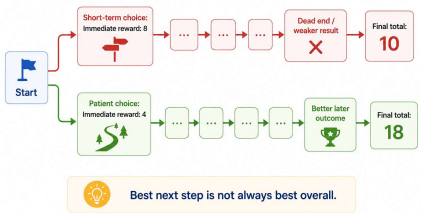
Greedy strategies can fail when:

- an early choice blocks a better later choice
- the local score hides a global tradeoff
- the rule does not match the real goal

Greedy needs evidence

41

The Risk of a Greedy Algorithm



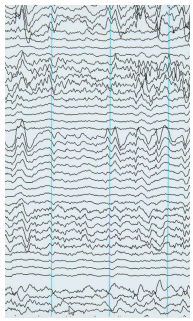
Greedy Risk

42

Strategy Family Comparison

Strategy	Basic Idea	Watch For
Brute force	try possibilities	too much work
Divide and conquer	split and combine	combine step
Dynamic programming	reuse saved results	identifying repeated subproblems
Greedy	best-looking next step	local choice may fail

43



PageRank As An Example

It considers:

- links between pages
- importance passed through relationships
- repeated updating of scores
- ranking based on structure

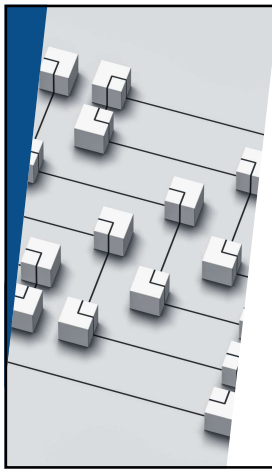
44

PageRank: What To Notice

Notice the algorithmic questions:

- What is represented?
- What is repeated?
- What score changes over time?
- What does the rank claim?
- What are the limits of that claim?

45



Linear Programming: *Recognition*

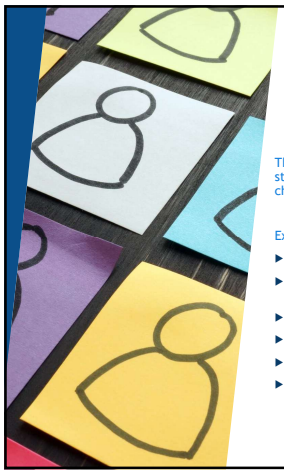
Linear programming helps optimize under constraints.

Plain-language frame:

- choose values
- follow constraints
- maximize or minimize a goal

For this course: recognize the idea.

46



Constraints Shape Strategy

The same problem can change strategy when constraints change.

Examples:

- exact answer required
- good-enough answer acceptable
- small data
- large data
- limited memory
- explanation required

47

Strategy Before Code

Before coding, describe:

- Strategy A
- Strategy B
- what each strategy assumes
- what evidence will compare them
- what would make one a better fit

48

Same Strategy, Different Form

Sometimes the strategy changes.

Sometimes the representation changes.

Sometimes both solutions solve the same task, but one fits the data shape more clearly.

49

Iteration and Recursion

50

Iteration

Iteration repeats steps using a loop.

Common signs:

- `for`
- `while`
- running total
- index or counter
- explicit work list or stack

51

Recursion

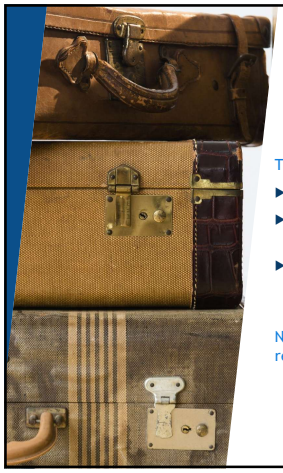
Recursion happens when a function solves a smaller version of the same problem.

It needs:

- a base case
- a recursive step
- progress toward stopping

52

Base Case



The base case answers:

- When are we done?
- What result can we return directly?
- How do we stop the repeated calls?

No base case means no reliable recursion.

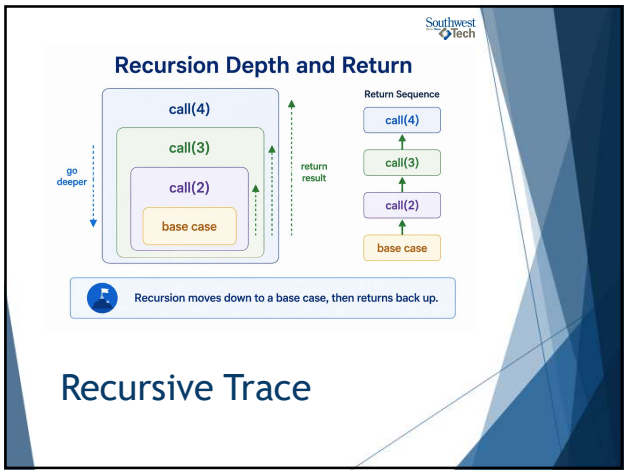
53

Recursive Trace

A recursive trace can show:

- call depth
- current input
- action taken
- returned result
- how totals combine

54

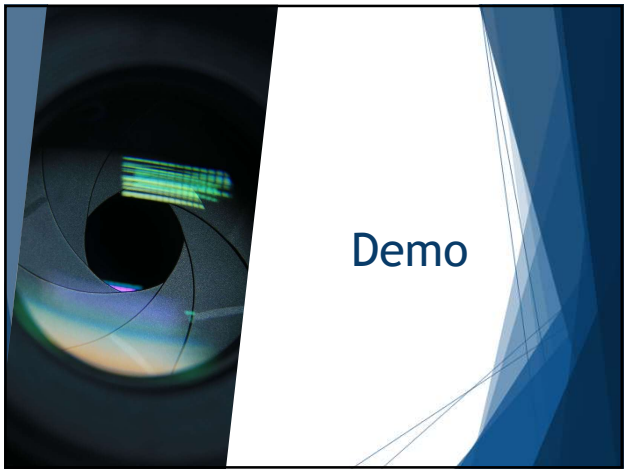


55

Iteration vs Recursion

Question	Iteration	Recursion
How does it repeat?	loop	function calls itself
What must be tracked?	loop state	call state and base case
When does it fit?	flat repeated work	nested or self-similar work

56



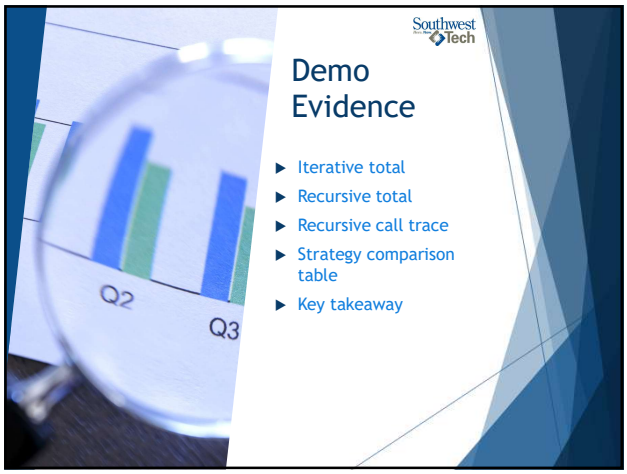
57

Demo Scenario

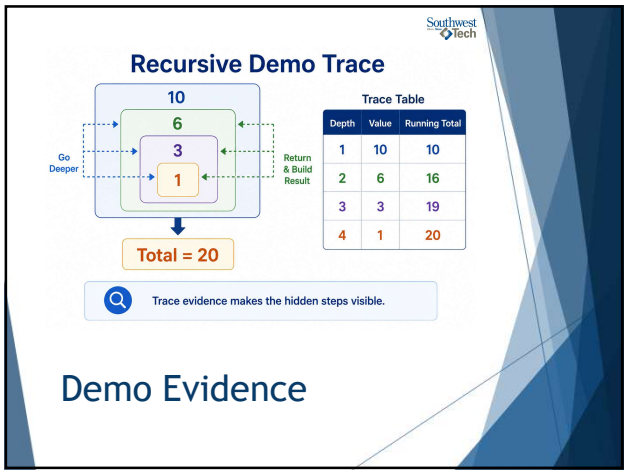
Problem:
Compare two strategies:
- iterative processing with an explicit work stack
- recursive processing of nested groups

Goal
Calculate the total value of nested donation envelopes

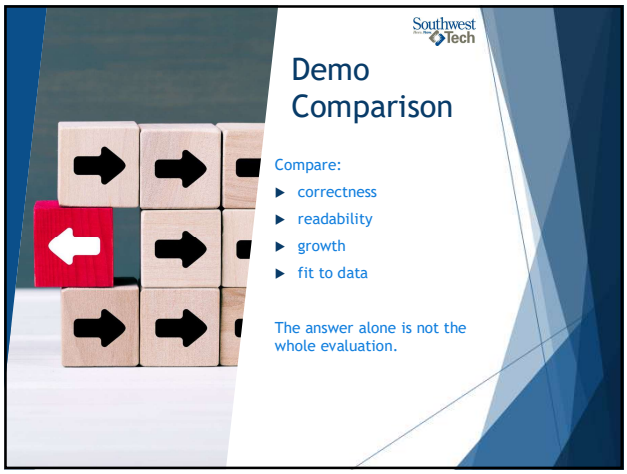
58



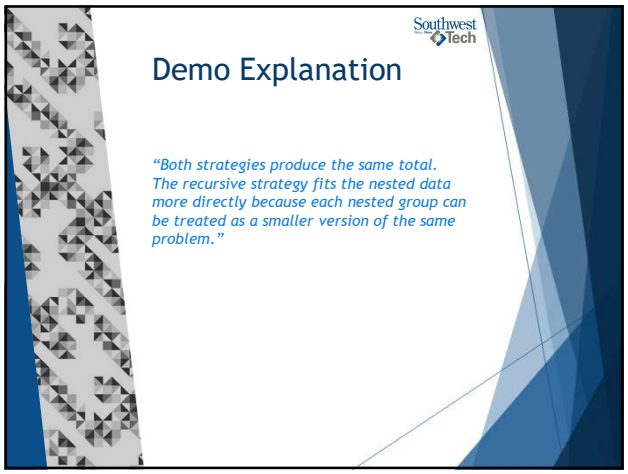
59



60

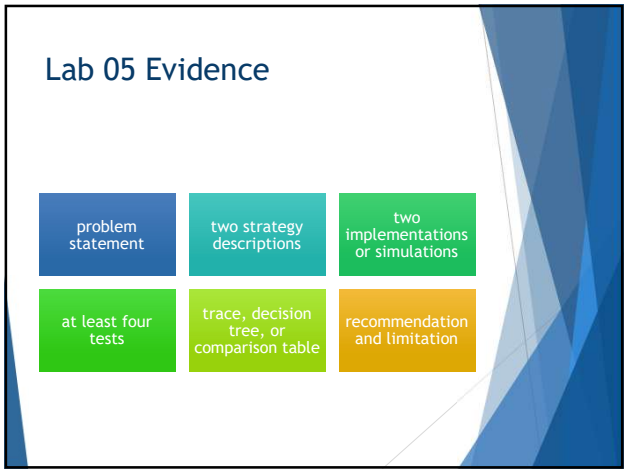


61

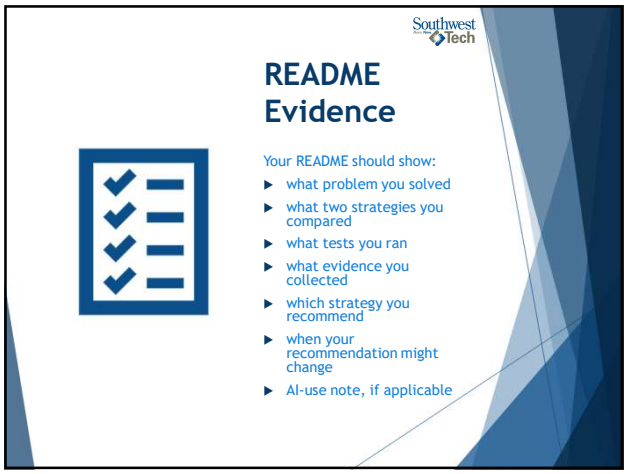


62

63



64



65

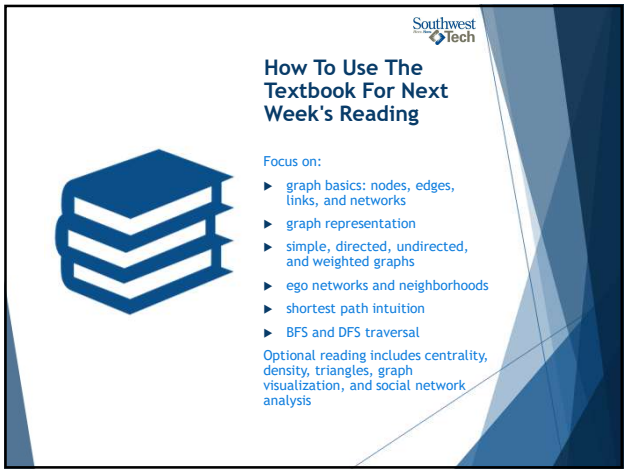
66


Southwest Tech
Wrap up

67


Southwest Tech
What To Carry Forward
Strategy choice is contextual.
Ask:
Does it produce the expected result?
Is it readable enough to explain?
How does the work grow?
Does it fit the data shape?
What tradeoff did I accept?

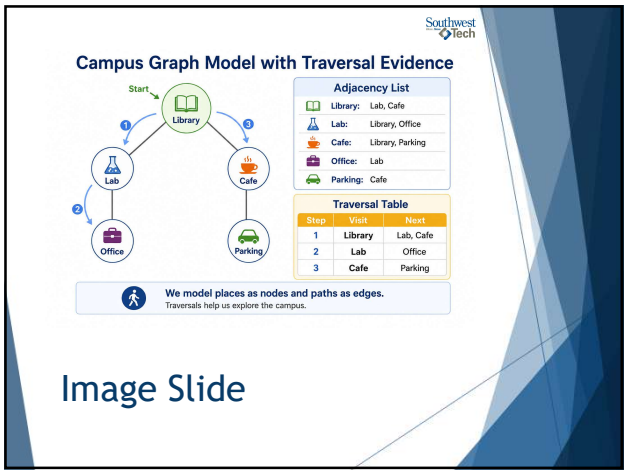
68


Southwest Tech
How To Use The Textbook For Next Week's Reading
Focus on:
graph basics: nodes, edges, links, and networks
graph representation
simple, directed, undirected, and weighted graphs
ego networks and neighborhoods
shortest path intuition
BFS and DFS traversal
Optional reading includes centrality, density, triangles, graph visualization, and social network analysis

69


Southwest Tech
Thank you!
Have Fun!

70


Southwest Tech
Campus Graph Model with Traversal Evidence
Start
Library
Lab
Cafe
Office
Parking
Adjacency List
Library: Lab, Cafe
Lab: Library, Office
Cafe: Library, Parking
Office: Lab
Parking: Cafe
Traversal Table
Step Visit Next
1 Library Lab, Cafe
2 Lab Office
3 Cafe Parking
We model places as nodes and paths as edges.
Traversals help us explore the campus.
Image Slide

71